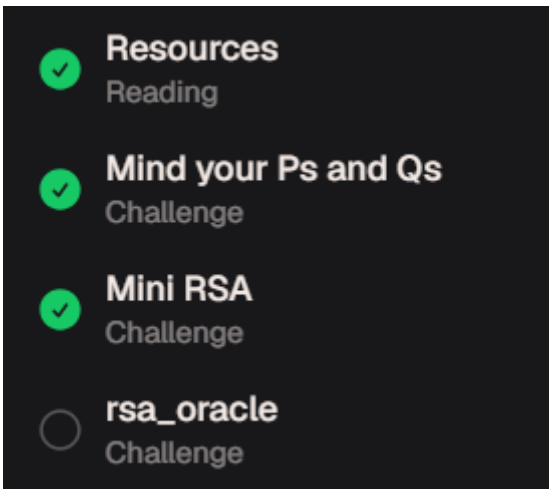
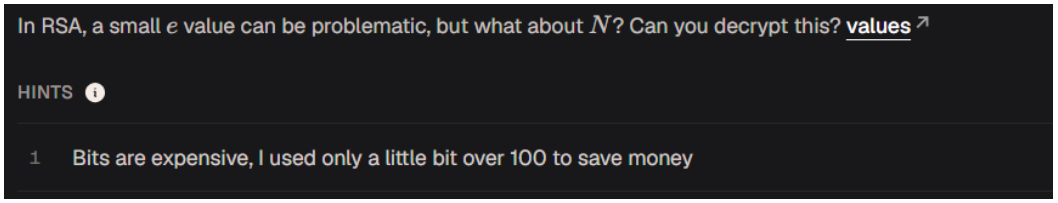
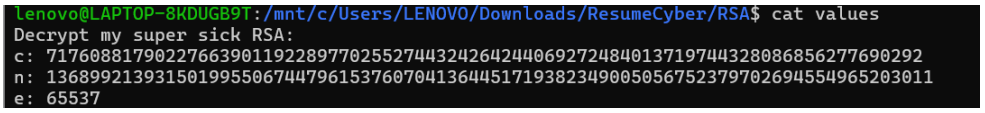


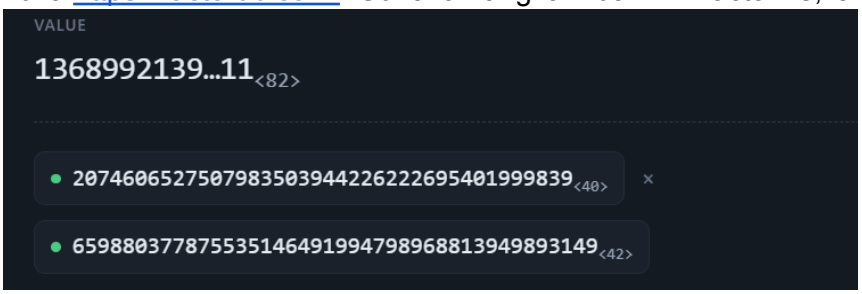
Challenge 1 : Mind your Ps and Qsr



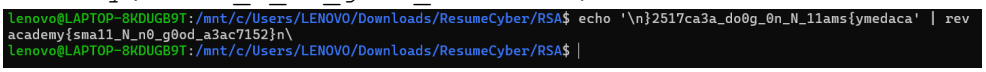
File Bernama : values

1. Kita lihat isi dari file.
2. 

3. Buka https://factordb.com/. Gunakan angka *n* dan klik factorize, lalu catat p dan q



4. Buat file solve bertipe python (**nano solve.py**). Source Code ada di file bernama solve.py
5. Jalankan file solve tersebut (**python3 solve.py**)
6. Ketemu flagnya akan tetapi dalam keadaan reverse yakni `b'\n}2517ca3a_do0g_0n_N_11ams{ymedaca'`
7. Tinggal kita balikan dan ketemu flagnya yakni `academy{sma11_N_n0_g0od_a3ac7152}`



Challenge 2 : Mini Rsa

What happens if you have a small exponent? There is a twist though, we padded the plaintext so that $(M * *e)$ is just barely larger than N. Let's decrypt this: **values**

HINTS

1. RSA **tutorial**
2. How could having too small of an e affect the security of this key?
3. Make sure you don't lose precision, the numbers are pretty big (besides the e value)
4. You shouldn't have to make *too* many guesses
5. **academy** is in the decrypted message, but not at the beginning

File bernama : values1

1. Kita lihat isi file tersebut.

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ cat values1
N: 16157656843214630540782260519598878842336783177348929017407633211352136367960754624019502746024050951385898980874283377
77584450132814889668660733557107718646717269919187065580712312669764271846738002252545316959285412725463851464957364202
61815693810544589811104967829354461491178200126099661909654163542661541699404839644035177445092988952614918424317082380
17438381902558507620664199347932657618079354432119435701891621511300974265440859708372450816921618200844969391722749781
31654444372201517541788989925461711067825681947947471001390843774746442699739386923285801022685451221261010798837646928
09227755619814566292469180303288004049276244256149776068993360178140161708660059348212746565539084136115402589067975751
14060456103104199255917164678161972735858939464790960448345988941481499050248673128656508055285037090026439683847266536
28316014207164301543481347346346973311218232867870670211605403661827750699766653456784676393869233506995575524443841537
79334400294983789553558775027432153057688148578644331512870
[*] N = 16157656843214630540782260519598878842336783177348929017407633211352136367960754624019502746024050951385898980874
28333775844501328148896686607335571077186467172699191870655807123126697642718467380022525453169592854127254638514649573
64202618156938105445898111049678293544614911782001260996619096541635426615416994048396440351774450929889526149184243170
82380174383819025585076206641993479326576180793544321194357018916215113009742654408597083724508169216182008449693917227
49781316544437220151754178898992546171106782568194794747100139084377474644269973938692328580102268545122126101079883764
69287755619814566292469180303288004049276244256149776068993360178140161708660059348212746565539084136115402589067975751
40604561031041992559171646781619727358589394647909604483459889414814990502486731286565080552850370900264396838472665362
83160142071643015434813473463469733112182328678706702116054036618277506997666534567846763938692335069955755244438415377933
44002949837895535587750274321530576881485786443315128 7
[*] e = 3
[*] c = 57097201750263178419445051663321827794197600311154322155634259018756200527916081633987502973042778864955375977
93006288112991932717211066611899319963701838062828209254458935983141207100322897221021063116398621664612418813778554568264
34043022583257841951018871627248541684702003178624049347421621997602223086988895202767821691757309664710332900774731822
69245461553908060851209149694528927423345871558880671158097412479425202494135508237765427141979769543939839645474158854944
97796731704194801876261983756511093491385632214811460508122022808238249641948118634778982733276631909976838533967548567
40987299287224914545024819209643705436617193628436345338680537047177323301194162660757103489761598885828533461196000569
74853545906914352354210842700313972898005708306666464115064482164735343062754084645726164346567909517470476669478575568
96756165696297419003719375390627511035337586115397485955619897923170604847178286434619891543704954877283984798455539174
95920818130003766392290200672893336450122809476169268494312

[+] k = 0
[+] Plaintext (bytes): b'
    academy{e_sh0u1d_b3_lArg3r_35f26e1a}\n'
[+] Plaintext (str):
    academy{e_sh0u1d_b3_lArg3r_35f26e1a}
```

2. Buat file solve bertipe python karena sudah ada bagian *n*, *e*, dan *c* (**nano solve1.py**). Source Code ada di file bernama solve.py
3. Jalankan file solve tersebut (**python3 solve1.py**)

```
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ python3 solve1.py
[*] N = 16157656843214630540782260519598878842336783177348929017407633211352136367960754624019502746024050951385898980874
28333775844501328148896686607335571077186467172699191870655807123126697642718467380022525453169592854127254638514649573
64202618156938105445898111049678293544614911782001260996619096541635426615416994048396440351774450929889526149184243170
82380174383819025585076206641993479326576180793544321194357018916215113009742654408597083724508169216182008449693917227
49781316544437220151754178898992546171106782568194794747100139084377474644269973938692328580102268545122126101079883764
69287755619814566292469180303288004049276244256149776068993360178140161708660059348212746565539084136115402589067975751
40604561031041992559171646781619727358589394647909604483459889414814990502486731286565080552850370900264396838472665362
83160142071643015434813473463469733112182328678706702116054036618277506997666534567846763938692335069955755244438415377933
44002949837895535587750274321530576881485786443315128 7
[*] e = 3
[*] c = 57097201750263178419445051663321827794197600311154322155634259018756200527916081633987502973042778864955375977
93006288112991932717211066611899319963701838062828209254458935983141207100322897221021063116398621664612418813778554568264
34043022583257841951018871627248541684702003178624049347421621997602223086988895202767821691757309664710332900774731822
69245461553908060851209149694528927423345871558880671158097412479425202494135508237765427141979769543939839645474158854944
97796731704194801876261983756511093491385632214811460508122022808238249641948118634778982733276631909976838533967548567
40987299287224914545024819209643705436617193628436345338680537047177323301194162660757103489761598885828533461196000569
74853545906914352354210842700313972898005708306666464115064482164735343062754084645726164346567909517470476669478575568
96756165696297419003719375390627511035337586115397485955619897923170604847178286434619891543704954877283984798455539174
95920818130003766392290200672893336450122809476169268494312

[+] k = 0
[+] Plaintext (bytes): b'
    academy{e_sh0u1d_b3_lArg3r_35f26e1a}\n'
[+] Plaintext (str):
    academy{e_sh0u1d_b3_lArg3r_35f26e1a}
```

4. Ketemu flagnya yakni `academy{e_sh0u1d_b3_lArg3r_35f26e1a}`

Challenge 3 : rca_oracle

Can you abuse the oracle?

An attacker was able to intercept communications between a bank and a fintech company. They managed to get the **message** (ciphertext) and the **password** that was used to encrypt the message. After some intensive reconassainance they found out that the bank has an oracle that was used to encrypt the password and can be found here **`nc xebec.cylabacademy.net 44130`**. Decrypt the password and use it to decrypt the message. The oracle can decrypt anything except the password.

INSTANCE

Expires in 14:31 Restart Instance

HINTS

1. Crytography Threat models: chosen plaintext attack.
2. OpenSSL can be used to decrypt the message. e.g **`openssl enc -aes-256-cbc -d ...`**
3. The key to getting the flag is by sending a custom message to the server by taking advantage of the RSA encryption algorithm.
4. Minimum requirements for a useful cryptosystem is CPA security.

File bernama : password.enc dan secret.enc

1. Lihat masing masing file tersebut

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ cat password.enc
34870802014302624395807016598518733470205107282829940878749375150711651620546337137894491822168096018634611739942757268947574790143352523235266600429586 06lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ cat secret.enc
Salted__Y◆◆kz)◆9◆ƚp2S.◆◆w@◆◆q◆◆◆G(◆n}◆◆a%◆×◆◆/◆◆◆◆i◆s]◆YXlenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$
```

2. Hubungkan dengan port yang disediakan yakni `nc chatelaine.cylabacademy.net 30317` (ga sesuai gambar karena waktunya habis)
3. Selanjutnya, kita ketik *e* dan ketik *1* agar muncul ciphertextnya. Ketika sudah tinggal kita exit dengan Ctrl + c.

```
*****************************************
***************THE ORACLE***************
*****************************************
what should we do for you?
E --> encrypt D --> decrypt.
e
enter text to encrypt (encoded length must be less than keysize): 1
1

encoded cleartext as Hex m: 31

ciphertext (m ^ e mod n) 43746717414118196530950652036383638397057601445241916336053581346841439783210958590471265856492728729087654320409430553992474997440703718104706823661006 89
```

4. Buat file solve bertipe python (**nano solve2.py**). Source Code ada di file bernama solve.py dan ada cipher yang udah diketahui.
5. Jalankan file solve tersebut (**python3 solve2.py)**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ python3 solve4.py
152548312172336050355150962113251172960874098303902458432792460122182909416090408684885616891700788123087668373226787562008841649087023614873571435160417786394059365645488103240255126909908593571458253792376452121450386197154720781616686386 07513383099220175229113878396403254923393466893597854185207255079534
```

6. Catat output tersebut karena merupakan ***blinded*** yang menjadi syarat langkah berikutnya
7. Sama seperti nomor 3, kita ketik *d* dan masukkan ***blinded*** agar muncul sebuah password. Ketika sudah tinggal kita exit dengan Ctrl + c.

```
what should we do for you?
E --> encrypt D --> decrypt.
d
Enter text to decrypt: 152548312172336050355150962113251172960874098303902458432792460122182909416090408684885616891700788123087668373226787562008841649087023614873571435160417786394059365645488103240255126909908593571458253792376452121450386197154720781616686386 07513383099220175229113878396403254923393466893597854185207255079534
decrypted ciphertext as hex (c ^ d mod n): 99c2fc19cb8
decrypted ciphertext:   /Á¸
```

8. Buat file solve bertipe python (**nano solve3.py**). Source Code ada di file bernama solve.py dan ada cipher yang udah diketahui.
9. Jalankan file solve tersebut (**python3 solve3.py**)

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/RSA$ python3 solve5.py
m = 10566420765880
password_int = 215641240120
password_bytes = b'25828'
password = 25828
```

10. Langkah terakhir OpenSSL: decrypt secret.enc dengan password yang sudah

diketahui. Querynya `openssl enc -aes-256-cbc -d -in secret.enc -pass pass:25828`

11. Ketemu flagnya yakni `academy{su((3ss_(r@ck1ng_r3@_2582816d}`
12. Langkahnya bisa diamati disini : https://chat.deepseek.com/share/nra1ae8030e71yrv7s